https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	r is the independent variable and *h* is the dependent variable **or** vary *r* and measure/determine *h*	**1**
	keep *P* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • labelled air blower • labelled ball vertically (by eye) above the blower • vertical (by eye) rule at least from top of blower to ball	**1**
	circuit diagram to determine *P*, e.g. voltmeter and ammeter connected to motor and power supply or wattmeter connected to motor and power supply	**1**
	use of micrometer/calipers/rule to measure diameter of ball	**1**
	measure the distance between the top/middle/bottom of the ball and the top of the blower	**1**
	Method of Analysis	
	plot a graph of *h* against $1 / r^3$ (or lg *h* against lg *r*)	**1**
	relationship valid if a straight line through (0,0) (for lg *h* against lg *r* straight line with gradient = –3)	**1**
	$K = \frac{4\pi g \times \text{gradient}}{3P}$ (for lg *h* against lg *r*, $K = \frac{4\pi g \times 10^{y\text{–intercept}}}{3P}$)	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	
	D1 use large box/tray to collect ball (to prevent ball rolling on floor/bouncing) **or** reasoned method to avoid draughts, e.g. switch off fans, close windows, use a screen	**6**
	D2 method to determine P from correct circuit, e.g. $P = I \times V$ or use wattmeter to measure P	
	D3 stand on bench with clamped rule vertically to measure vertical distance	
	D4 method to ensure clamped rule to measure h is vertical, e.g. correctly positioned set square indicated at right angles between the rule and the horizontal surface **or** plumb line shown in appropriate position	
	D5 $r = d / 2$ when diameter measured	
	D6 repeat diameter measurement in different directions and find average	
	D7 repeat experiment for each value of r and determine average h	
	D8 method to determine h, e.g. h = reading top of ball – r – top of blower **or** h = reading bottom of ball + r – top of blower **or** h = (distance from top of ball to top of blower + distance from bottom of ball to top of blower) / 2	
	D9 wait for the ball to become stationary (vertically)	
	D10 video (camera) shown level with elevated ball and description of playback frame-by-frame or slow motion	

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